🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Use Ratio Reasoning

Lesson 3-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to solve problems with proportions and scaling.
Language Objective: I can explain my reasoning using the words proportion, scale, equivalent, and unit rate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Proportion | Cross-multiply | Scale | Equivalent | Unit rate

👆 *Tap any word to see what it means and a picture.*

1. An equation that states two ratios are equal is a ________.
2. Multiplying the numerator of one ratio by the denominator of the other to check a proportion is to ________.
3. To multiply a ratio up or down by the same factor is to ________ it.
4. Two ratios that show the same comparison are ________.
5. The amount for exactly one unit is the ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

You sorted ratio pairs into equivalent and not equivalent. How does ratio reasoning, like cross-multiplying, prove that 4:7 and 12:21 are equivalent?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Proportion** — A math sentence saying two ratios are equal.
  Proporción — Una oración matemática que dice que dos razones son iguales.
- [ ] **Cross-multiply** — Multiplying across two ratios to check if they are equal.
  Multiplicación cruzada — Multiplicar en cruz dos razones para ver si son iguales.
- [ ] **Scale** — To multiply or divide both parts of a ratio by the same number.
  Escalar — Multiplicar o dividir ambas partes de una razón por el mismo número.
- [ ] **Equivalent** — Having the same value.
  Equivalente — Tener el mismo valor.
- [ ] **Unit rate** — A rate for just 1 of something. You find it by dividing.
  Tasa unitaria — Una tasa para solo 1 de algo. La encuentras al dividir.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I reasoned that 4:7 and 12:21 are ____ because ____.

Razoné que 4:7 y 12:21 son ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.

Evidence: Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Reasoning: This evidence connects the definition of proportion to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I reasoned that 4:7 and 12:21 are ____ because ____.**
  Razoné que 4:7 y 12:21 son ____ porque ____.
- **Cross-multiplying gives ____ and ____, which are ____.**
  La multiplicación cruzada da ____ y ____, que son ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Proportion
2. **Notes 2:** Cross-multiply
3. **Notes 3:** Scale
4. **Notes 4:** Equivalent
5. **Notes 5:** Unit rate
6. **Watch for this mistake:** *A common mistake in Use Ratio Reasoning is scaling by adding instead of multiplying: since the banquet has 120 − 8 = 112 more guests than the recipe serves, some students add 112 to the ingredient amount instead of finding the scale factor. For Chef Reyes's recipe that serves 8 people with 5 cups of tomatoes, scaling up to 120 guests this way gives 5 + 112 = 117 cups of tomatoes — a wildly wrong answer. Ratios scale multiplicatively, not additively: find the scale factor first (120 ÷ 8 = 15), then multiply the ingredient by that same scale factor: 5 × 15 = 75 cups of tomatoes. Before you submit, check that you multiplied every part of the ratio by the same scale factor instead of adding the same amount to each.*
7. **Try It 1:** A. 12 cups *— Scale factor: 18 servings divided by 6 servings = 3. Multiply the tomatoes by 3: 4 x 3 = 12 cups.*
8. **Try It 2:** A. 12 dishes *— Scale factor: 20 minutes divided by 5 minutes = 4. Multiply the dishes by 4: 3 x 4 = 12 dishes.*